

Worked for 39s

A7 — Clearance Progression

This step changes one assumption from our earlier rough planning:

Phantom Q should not use one simple numerical clearance level such as “Level 1 → Level 2 → Level 3”.

Now that A1–A6 have established **two different progression branches**—security operations and engineering/quantum—a single number would create bad logic.

For example:

```
Player earns Red Team access
      ↓
Clearance increases from 5 → 6
      ↓
system accidentally implies
Quantum access is now allowed
```

That makes no organisational sense.

Instead, Phantom Q should use a **credential + capability + mission-authorisation model**.

A7.1 The Phantom Q access model

Every player/NPC has an **access profile**.

```
PHANTOM Q ACCESS PROFILE
|
|— IDENTITY
|
|   Is this person recognised?
|
|— BASE ROLE
|
|   Trainee / Analyst / Engineer / Red Team / etc.
|
|— ZONE ACCESS
|
|   Which physical areas may they enter?
|
|— CAPABILITY ENDORSEMENTS
|
|   What equipment/functions may they use?
|
|— MISSION AUTHORISATIONS
|
|   What temporary areas/actions are allowed?
|
|— HIGH-RISK ACTION AUTHORITY
|
|   Requires explicit signed approval
```

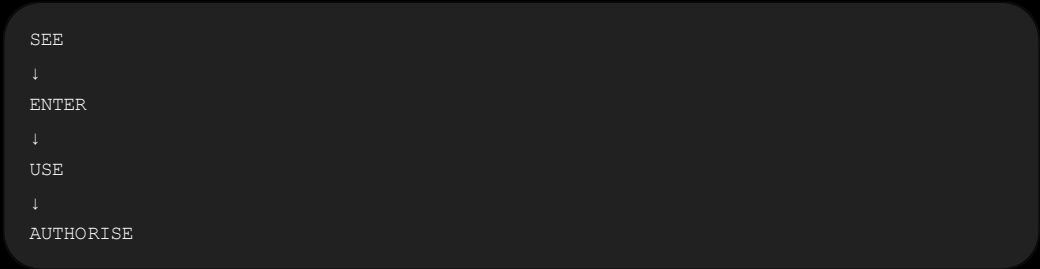
That is substantially closer to a professional access-control architecture.

A7.2 Entry is not the same as permission to operate

This distinction is critical.

A player might be allowed to enter SOC without being allowed to operate every SOC console.

So we separate:



These are different permissions.

For example:

SOC		
SEE		✓
ENTER		✓
USE TRAINING CONSOLE		✓
USE CASE PQ-004		✓
ADMINISTER SOC		✗
AUTHORISE ATTACK		✗

This will make the Headquarters much more believable.

A7.3 The five access states

For the environment, I recommend five simple player-facing states.

State	Meaning
OPEN	Normal authorised access
LIMITED	Enter, but only designated equipment/functions
MISSION	Temporary case/mission-specific access
VISIBLE / LOCKED	Player can see the capability but cannot enter
RESTRICTED	No current access; minimal information exposed

This is better than constantly displaying numeric levels.

A7.4 The player's initial credential

At the very beginning:

PHANTOM Q TRAINING CREDENTIAL

Identity
UNVERIFIED

Role
TRAINEE

Operational access
NONE

Specialist endorsements
NONE

Mission authority
NONE

After Registration + Eve's authorised credential check:

IDENTITY
VERIFIED ✓

ROLE
PHANTOM Q TRAINEE

HQ ENTRY
APPROVED ✓

CENTRAL OPERATIONS
LIMITED ✓

This is the first meaningful progression event.

A7.5 Prologue access state

Once the player passes T0:

ENTRANCE / RECEPTION	OPEN
CENTRAL OPERATIONS	LIMITED
FOUNDATIONS	VISIBLE / LOCKED
COMMUNICATIONS	VISIBLE / LOCKED
CRYPTOGRAPHY	LOCKED
SOC	VISIBLE / RESTRICTED
RED TEAM	RESTRICTED
ENGINEERING	VISIBLE / LOCKED
QUANTUM	RESTRICTED
ADVANCED COMPUTE	RESTRICTED
SECURE CORE	RESTRICTED

Central Operations is **limited** because the trainee can use:

```
Workstation 04
basic operational tools
approved communication interface
```

but cannot use:

```
HQ Command
incident authority controls
administrative systems
```

That distinction is important from the start.

A7.6 Central Operations has an internal access hierarchy

Central Operations itself contains two access domains.

```
CENTRAL OPERATIONS
|
|— GENERAL OPERATIONS FLOOR
|
|   Trainee-accessible
|
|— COMMAND / AUTHORISATION ZONE
    Restricted
```

So the learner may physically spend a large part of the campaign in Central Ops without ever having the authority to issue HQ commands.

Incident 06 therefore still makes sense:

HQ signs the offensive authorisation.

The player does not become HQ simply because they became competent.

A7.7 Prologue → PQ-001

When PQ-001 is created, the player does **not immediately receive broad security privileges**.

Instead:

```
PQ-001 CREATED

FOUNDATIONS TRAINING
AUTHORISED ✓
```

The Foundations door changes:

```
BEFORE

FOUNDATIONS
TRAINING CLEARANCE REQUIRED
```

AFTER

FOUNDATIONS

ACCESS GRANTED

This is the player's first visible HQ unlock.

A7.8 Incident 01 — Foundations endorsement

During Incident 01:

ZONE ACCESS

Foundations ✓

CAPABILITIES

Timeline reconstruction

Evidence classification

CIA assessment

Identity comparison

Verification logic

At successful completion, the trainee earns a persistent capability:

Foundations / Security Reasoning Endorsement

Conceptually:

PQ-CAP-FND

Security Foundations

VERIFIED ✓

We do not need to expose that code to the player, but internally this is useful.

A7.9 Incident 01 → Incident 02

Completion of Foundations proves enough capability to access the Cryptography Lab for the next training stage.

So:

CRYPTOGRAPHY LAB

TRAINING ACCESS GRANTED

But this does **not** automatically grant:

Advanced Compute

Quantum

SOC

Red Team

This is where our non-linear access system becomes valuable.

A7.10 Incident 02 — Cryptography access

At the beginning of Incident 02:

CRYPTOGRAPHY LAB
OPEN / TRAINING

COMMUNICATIONS
LIMITED / MISSION

SOC
RESTRICTED

Inside Crypto, the learner may use:

plaintext workbench
A7 training key station
encryption transformation
training key-transfer evaluation

But not:

production key authority
HQ identity keys
administrative crypto infrastructure

Again:

Entering a room does not make the trainee its administrator.

A7.11 Communications should initially be capability-limited

When Incident 02 reaches ciphertext transmission, the player receives:

COMMUNICATIONS
MISSION ACCESS

PURPOSE
Protected training transfer

The learner can use:

Alice/Bob training link
message-send interface
session display

but not:

network administration
system configuration
other users' communications

This is both realistic and useful for role-based gameplay.

A7.12 Incident 03 — Communication capability becomes established

Once the learner:

```
verifies Bob
authenticates Bob
confirms authorisation
releases A7 correctly
successfully decrypts
completes the R100 workflow
```

they have demonstrated an operational capability.

At that point I would make:

```
CRYPTO TRAINING ACCESS      PERSISTENT ✓
COMMUNICATIONS TRAINING ACCESS  PERSISTENT ✓
```

for the current campaign.

That does **not** mean unrestricted administration.

It means the learner has proven enough competence to use the approved Phantom Q training systems.

A7.13 Incident 04 — SOC should be mission-gated first

This is where access progression should become more serious.

When the R1 000 diversion occurs, SOC does not simply unlock because:

“You reached Level 4.”

Instead:

```
CASE
PQ-004

INCIDENT ESCALATION
CONFIRMED

SOC CASE ACCESS
REQUESTED
```

Once the communication evidence supports escalation:

```
PQ-004
SOC INVESTIGATION ACCESS
GRANTED
```

So entry is initially:

MISSION

not permanent unrestricted SOC access.

That is much more believable.

A7.14 SOC access in Incident 04

The learner can:

ENTER SOC	✓
VIEW PQ-004 EVIDENCE	✓
USE TIMELINE	✓
USE SESSION ANALYSIS	✓
USE CASE CORRELATION	✓
PRESERVE PQ-004 RECORDS	✓

But cannot:

VIEW UNRELATED CASES	✗
ADMINISTER SOC	✗
CHANGE SECURITY POLICY	✗
AUTHORISE OFFENSIVE ACTION	✗

This is a very important access boundary.

A7.15 End of Incident 04

The learner has now demonstrated legitimate incident-analysis ability.

So I would award:

Incident Investigation Endorsement

This means future authorised investigations do not require them to repeat basic SOC training.

But active case access still depends on a case.

Therefore:

CAPABILITY
SOC Investigation ✓
CASE ACCESS
requires active authorised case

This is a strong model.

A7.16 Incident 05 — Red Team becomes visible operationally

At Incident 05 start:

```
SOC
PQ-004 ACCESS ✓

CRYPTO
ACCESS ✓

COMMUNICATIONS
ACCESS ✓

RED TEAM
VISIBLE / LOCKED
```

Then:

```
Scene 05
RED TEAM GOES ON STANDBY
```

The state changes to:

```
RED TEAM OPERATIONS

UNIT 01
STANDBY

PLAYER ACCESS
NOT AUTHORISED
```

That is deliberate.

The learner can help **prepare** Red Team without being allowed inside.

A7.17 Why Red Team must remain locked through Incident 05

Incident 05 is about:

```
BUILD TRUST
↓
ESTABLISH COMMAND AUTHORITY
↓
CREATE S1
↓
PREPARE OPERATION
```

Not:

```
become offensive operator immediately
```

So the physical Red Team boundary is part of the lesson.

The learner must first prove:

```
trusted command
signature verification
fresh session establishment
```

before that door becomes available.

A7.18 Incident 05 end — Red Team mission grant

Scene 28 finally produces:

```
PQ-004 RECOVERY OPERATION
READY
```

Now the player credential receives:

```
MISSION AUTHORISATION
```

```
Case:
```

```
PQ-004
```

```
Zone:
```

```
Red Team Operations
```

```
Purpose:
```

```
Recovery Operation
```

```
Status:
```

```
ACTIVE
```

Then T2 changes:

```
RED TEAM OPERATIONS
```

```
MISSION ACCESS
```

```
GRANTED
```

This is one of the strongest physical progression moments in the game.

A7.19 But Red Team access is still not attack authority

This distinction is absolutely critical.

Entering Red Team gives the player permission to **participate in the operation**.

It does **not** give permission to attack arbitrary systems.

So:

```
ENTER RED TEAM
```

```
✓
```

```
USE PQ-004 OPERATIONAL CONSOLES
```

```
✓
```

```
INVESTIGATE PQ-004
```

```
✓
```

ATTACK ANY TARGET
×

The latter requires another authorisation layer.

A7.20 Incident 06 — target-specific offensive authority

The sequence remains:

```
MISSION AUTHORITY
PQ-004
    ↓
allows investigation
    ↓
TARGET EVIDENCE
10.91.24.18
    ↓
REQUEST OFFENSIVE AUTHORITY
    ↓
HQ SIGNED APPROVAL
    ↓
VERIFY
    ↓
TARGET + SERVICE + ACTION
MATCH
    ↓
ENGAGE
```

This should be represented as a **temporary action grant**.

For example:

```
OFFENSIVE AUTHORISATION

Case
PQ-004

Target
10.91.24.18

Service
Remote Management

Action
Controlled Access

Status
VERIFIED

Other targets
NOT AUTHORISED

Other services
OUT OF SCOPE
```

This is much stronger than a generic:

Red Team clearance = can hack.

A7.21 High-risk authority is deliberately narrow

We should adopt a general Phantom Q rule:

The more consequential the action, the narrower the authorisation.

For example:

```
GENERAL HQ ENTRY
broad

CRYPTO TRAINING ACCESS
domain-specific

PQ-004 SOC ACCESS
case-specific

RED TEAM ACCESS
mission-specific

OFFENSIVE ACTION
target + service + action specific

FINANCIAL RECOVERY
case + amount + destination specific
```

This is a very good security concept to teach indirectly.

A7.22 Financial recovery authority

Incident 06 later requires:

```
R1 000 recovery
```

That should have its own authority.

Not because the learner is Red Team.

HQ sends:

```
PQ-004 FINANCIAL RECOVERY APPROVAL

Case
PQ-004

Amount
R1 000

Source
....1058

Destination
PQ Secure Recovery
```

The payment system verifies the signed authorisation.

So even at the end of the game:

```
RED TEAM ACCESS
#
FINANCIAL AUTHORITY
```

Excellent separation of duties.

A7.23 What happens after Incident 06?

We should distinguish:

Skill earned

The player has proven:

```
Incident investigation
Cryptographic workflow
Trusted command verification
Red Team operational participation
Evidence preservation
Recovery operation
```

Those skills remain.

Mission authority

```
PQ-004 offensive authorisation
```

expires when the case closes.

Case-specific system access

```
PQ-004 target access
financial recovery authority
```

also expires.

That means completing the mission doesn't permanently leave dangerous controls open.

A7.24 Red Team after completion

I would not completely relock Red Team as though the player learned nothing.

Instead:

```
RED TEAM OPERATIONS

PLAYER STATUS
Operationally Qualified / Approved Participant

GENERAL TRAINING ACCESS
AVAILABLE ✓

LIVE OFFENSIVE ACTION
REQUIRES MISSION AUTHORITY
```

That gives progression without breaking professional security logic.

A7.25 Current cyber campaign should NOT unlock Engineering automatically

This is where our branch-based design becomes essential.

At the end of Incident 06:

```
CYBER / SECURITY CAPABILITY
significantly developed ✓
```

But:

```
ENGINEERING ENDORSEMENT
not necessarily earned

QUANTUM LAB
not automatically unlocked

ADVANCED COMPUTE
not automatically unlocked
```

Because:

Cyber competence is not the same thing as laboratory engineering competence.

This is much more defensible for an engineering training platform.

A7.26 Engineering progression should become its own branch

Later we could have:

```
ENGINEERING PATH

Basic Technical Access
↓
Instrumentation
↓
Embedded Systems
↓
Hardware Integration
↓
QKD Equipment
↓
Quantum Laboratory
```

while the security path is:

```
SECURITY PATH

Foundations
↓
Crypto
```

↓
Communications
↓
Incident Investigation
↓
Red Team Operations

Eventually the two converge.

That is probably where **Physical QKD** becomes extremely powerful.

A7.27 Physical QKD can require dual competence

For example, entry into the full Quantum practical could eventually require:

```
CRYPTO / SECURITY ENDORSEMENT
  ✓
  +
ENGINEERING / HARDWARE ENDORSEMENT
  ✓
  ↓
QUANTUM PRACTICAL ACCESS
```

This gives the Headquarters progression real depth.

And it prevents Quantum from simply being:

“the next level after Red Team.”

A7.28 Advanced Compute can similarly have multiple entry routes

For example:

```
CRYPTO SPECIALIST
  \
  \
    ADVANCED COMPUTE
  /
  /
SOC ANALYTICS

or

QUANTUM SIMULATION
```

Different learners may eventually unlock it through different capabilities.

Our architecture already supports this.

A7.29 Secure Core should require combinations, not one achievement

The Secure Core should be fundamentally different.

Not:

```
Reach Level 7
↓
Secure Core unlocked
```

Instead something more like:

```
IDENTITY VERIFIED
+
HIGH-TRUST ROLE
+
REQUIRED DOMAIN ENDORSEMENTS
+
CURRENT MISSION NEED
+
SPECIFIC AUTHORISATION
↓
SECURE CORE ACCESS
```

This keeps it valuable for future Acts.

A7.30 Player-facing credential card

This system can become a meaningful part of the UI.

Instead of a generic XP level:

```
PHANTOM Q CREDENTIAL
_____

ROLE
Trainee Security Engineer

VERIFIED CAPABILITIES
✓ Security Foundations
✓ Cryptography — Symmetric
✓ Communications Trust
✓ Incident Investigation
✓ Asymmetric Trust
✓ Evidence Handling
✓ Recovery Operations

ZONE ACCESS
Central Operations
Foundations
Cryptography
Communications
SOC — authorised cases
Red Team — operational/training

ACTIVE MISSION
None

RESTRICTED
Engineering
Quantum
Advanced Compute
Secure Core
```


That is far more meaningful than:

```
LEVEL 6  
2450 XP
```

We can still have progression scoring elsewhere, but **access itself should remain professionally structured**.

A7.31 Door behaviour

When the player approaches a door, the system checks:

```
IDENTITY  
  ↓  
ZONE PERMISSION  
  ↓  
REQUIRED CAPABILITY  
  ↓  
MISSION GRANT if required  
  ↓  
ACCESS DECISION
```

Conceptually:

```
CanEnterZone =  
  IdentityVerified  
  AND  
  (  
    PermanentZonePermission  
    OR ActiveMissionGrant  
  )  
  AND RequiredCapabilitySatisfied
```

For high-risk areas we add additional conditions.

A7.32 Player-facing denial should explain the reason

Do not just display:

```
ACCESS DENIED
```

Use meaningful feedback.

Examples:

```
CRYPTOGRAPHY LAB  
  
ACCESS REQUIRES  
Security Foundations
```

or:

```
SOC
```

SAFE AUTHORIZATION REQUIRED

or:

RED TEAM OPERATIONS

MISSION AUTHORISATION REQUIRED

or:

QUANTUM LAB

SPECIALIST TECHNICAL CLEARANCE REQUIRED

For Secure Core, we can reveal less:

SECURE CORE

RESTRICTED

because the player doesn't necessarily need to know all requirements early.

A7.33 Access-reader visual language

One consistent reader system:

```
WHITE / IDLE
Credential not scanned

BLUE
Checking

GREEN
Access granted

AMBER
Limited / mission-specific

RED
Access denied
```

We can later choose the exact visual palette consistent with Phantom Q branding, but the behaviour should remain consistent.

A7.34 Doors should remember mission state

Example T2 Red Team door:

Prologue—Incident 04

RESTRICTED

Incident 05

RED TEAM
STANDBY

PLAYER
NOT AUTHORISED

Incident 05 Scene 28

PQ-004 MISSION ACCESS
GRANTED

Incident 06

ACTIVE MISSION
PQ-004

After Incident 06

TRAINING / QUALIFIED ACCESS
AVAILABLE

LIVE OPERATIONS
MISSION AUTHORITY REQUIRED

Same door.

Different world state.

A7.35 NPCs use exactly the same system

This is important from A6.

Alice:

Central Ops	✓
Communications	✓
Crypto	✓
Red Team	×

Eve:

Identity / Security	✓
Communications	✓
SOC	✓
Red Team	normally ✗

Red Team Unit 01:

SOC	✓
Red Team	✓
Crypto	limited as required
Secure Core	not implied

Engineering staff:

Engineering	✓
Service Spine	✓
Quantum	role-dependent
Red Team	×

The player sees that **access policy applies to the whole organisation**, not just them.

A7.36 Character clearance can tell stories without dialogue

Imagine the player sees:

```

Quantum specialist
↓
badge reader
↓
Quantum gate opens
↓
player approaches
↓
ACCESS RESTRICTED

```

No tutorial paragraph is required.

The building teaches:

That person has a capability I have not yet earned.

That is exactly the kind of low-reading environmental learning we want.

A7.37 Visibility progression

Clearance should also affect **what the player can see**, not only where they can walk.

I would use three visibility categories.

Transparent / preview

Examples:

```

Foundations
SOC
Engineering

```

Player can see enough to understand their existence.

Controlled preview

Examples:

```

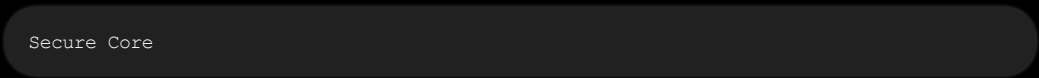
Red Team
Quantum
Advanced Compute

```

Partial sightlines, secure glazing, limited interior exposure.

Concealed

Example:

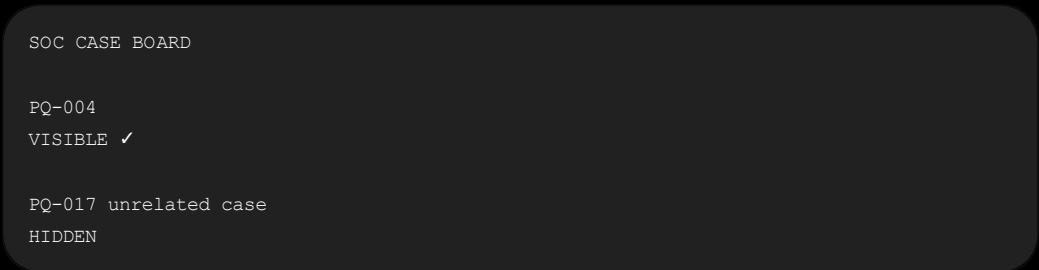


The player knows it exists but gains little operational detail.

This makes deeper security feel meaningful.

A7.38 Information access follows the same pattern

Even inside an authorised room:



This matches our existing server principle:

Capability can be universal; live information remains role-specific.

Physical access and information access are separate controls.

A7.39 Clearance progression through the current campaign

Here is the condensed campaign state.

Campaign state	New access/capability
Arrival	Reception only
Identity complete	Limited Central Ops
PQ-001 opened	Foundations access
Incident 01 complete	Foundations endorsement + Crypto training access
Incident 02	Mission-limited Communications access
Incident 03 complete	Crypto + Communications operational training capability
Incident 04 escalation	PQ-004 SOC mission access
Incident 04 complete	Incident Investigation endorsement

Campaign state	New access/capability
Incident 05	Asymmetric/trusted-command capability; Red Team visible on standby
Incident 05 Scene 28	PQ-004 Red Team mission access
Incident 06 target validated	May request target-specific offensive authority
Incident 06 authorised	Controlled target/service/action access only
Incident 06 recovery stage	Signed financial recovery authority only for PQ-004
Incident 06 complete	Recovery Operations capability retained; temporary authorities expire

That is the progression I would use.

A7.40 Zone matrix across the current campaign

Legend:

O = Open
L = Limited
M = Mission-specific
V = Visible / locked
R = Restricted

Zone	Prologue	I01	I02	I03	I04	I05
Reception / ID	O	O	O	O	O	O
Central Ops	L	L	O	O	O	O
Foundations	V→O	O	O	O	O	O
Crypto	R	V→O	O	O	O	O
Communications	V	V	M	O	O	O
SOC	V/R	V/R	R	R	M	M
Red Team	R	R	R	R	V/R	V
Engineering	V/R	V/R	V/R	V/R	V/R	V/R
Quantum	R	R	R	R	R	R
Advanced Compute	R	R	R	R	R	R
Secure Core	R	R	R	R	R	R

This makes the current campaign very clear.

A7.41 This exposes one good refinement to A5

During Incident 04–06, SOC should remain **case-authorised**, not simply become an unrestricted permanent player area.

So when the player crosses T1:

```
CASE
PQ-004

SOC ACCESS
ACTIVE
```

That reinforces the idea that the learner is working an authorised incident.

A5 does not need a route change.

Only the access-state logic becomes more precise.

A7.42 This exposes another refinement to A4

The Technical Gate T3 should exist and be visible in Release 1 even though the player cannot yet use it.

That means from the Foundations/Central Ops side the player can see:

```
PHANTOM Q ENGINEERING

SPECIALIST ACCESS REQUIRED
```

and beyond that perhaps signs for:

```
QUANTUM WING
```

This helps establish the larger Phantom Q world early.

A7.43 Emergency/life-safety exception

Since we want the architecture to have industrial credibility, one distinction should exist in the technical system:

Security access restrictions must not be confused with emergency egress.

Normal entry may be restricted:

```
Red Team entry
DENIED
```

but emergency evacuation routes must still work according to the eventual building/safety design.

We do not need to make this a major gameplay mechanic now, but we should not create an access-control architecture that implies people are physically trapped during an

emergency.

A7.44 Unity architecture implication

Later, I would not create:

```
Player.ClaranceLevel = 5
```

as the main system.

Instead something conceptually closer to:

```
AccessProfile
|
|— VerifiedIdentity
|— BaseRole
|— ZonePermissions[]
|— CapabilityEndorsements[]
|— ActiveMissionGrants[]
|— ActiveActionAuthorisations[]
```

And a door asks the access service:

```
Can this identity
enter this zone
for this purpose
right now?
```

That is much more future-proof.

A7.45 Loop against A1–A6

A1 — Building meaning: stronger. Access now genuinely represents competence, trust and responsibility.

A2 — Spatial hierarchy: passes. The increasing security depth remains meaningful without forcing a single linear clearance ladder.

A3 — Adjacency: passes. Direct adjacency does not create a security bypass because transitions still enforce policy.

A4 — Floor plan: passes. No room needs moving; T3 becomes more important as a visible future gateway.

A5 — Campaign routes: passes with the refinement that SOC and Red Team access are case/mission-specific rather than blanket unlocks.

A6 — Character routes: passes. NPCs can use the same access architecture with role-specific permissions.

A7 — Canonical lock

The main thing I would now lock is:

Phantom Q does not use a single linear clearance number.

The canonical model is:

```
IDENTITY
+
ROLE
+
PERMANENT CAPABILITY
+
ZONE PERMISSION
+
CURRENT MISSION
+
SPECIFIC ACTION AUTHORITY
```

And the progression philosophy becomes:

```
LEARN
↓
DEMONSTRATE
↓
EARN CAPABILITY
↓
GAIN APPROPRIATE ACCESS
↓
MISSION CREATES TEMPORARY AUTHORITY
↓
HIGH-RISK ACTION REQUIRES
EXPLICIT VERIFIED APPROVAL
```

That gives us a system that works for both **game progression** and **credible professional security practice**.

The next piece, **A8 — Interaction Distribution**, should now map the whole HQ again and determine **what the player physically does in each room**: which objects are interactive, which interactions are permanent versus incident-specific, where room-entry pop-ups belong, what information appears on terminals versus physical objects, and how we prevent every room from degenerating into “walk to monitor and click”.